

ASHWIN RAJAN R

Backend Engineer · Python & AI Infrastructure

ashwinrajan123123@gmail.com | +91-7010704499 | [linkedin.com/in/ashwin-rajan-rajamanickam](https://www.linkedin.com/in/ashwin-rajan-rajamanickam) | github.com/ashwinrajan-159

SUMMARY

Backend Engineer focused on scalable, low-latency Python systems using FastAPI, async programming, PostgreSQL, Redis, Docker, and cloud-native tooling. Experienced building production REST APIs, distributed systems, event-driven pipelines, and AI-powered applications on AWS, with interest in ML infrastructure and model serving.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/Node.js, C++, SQL, Bash

Backend: FastAPI, AsyncIO, REST APIs, OpenAPI/Swagger, async SQLAlchemy, Celery, WebSockets, JWT Authentication, OAuth2, Rate Limiting, Background Tasks, gRPC (learning), Nginx

Cloud & DevOps: AWS (EC2, ECS, S3, IAM, VPC), Azure, Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Linux, cloud-native deployments

Data & Messaging: PostgreSQL, Redis, Redis Streams, Kafka, MinIO/S3, FAISS

AI / ML: LLMs, embeddings, vector search, RAG, model serving / ML inference, OCR, SHAP explainability

Systems & Practices: Distributed systems, event-driven architecture, microservices, fault tolerance, Git, TCP/IP, DNS, VPC, pytest, structured logging, health checks

EXPERIENCE

ZettaOne — *Software Engineer Intern, Backend / Cloud Infrastructure*

Bengaluru | Jun 2026 – Present

- Developed asynchronous FastAPI services for AWS-hosted IoT telemetry ingestion and ML inference workloads, processing edge-device data in near-real time.
- Optimized backend API performance by profiling pipeline bottlenecks and reducing request-processing latency.
- Built Dockerized, cloud-native inference services deployed into containerized AWS environments and CI/CD workflows.

PROJECTS

TrustLens — *AI-Powered Explainable Fraud Detection Platform*

GitHub | 2026

- Built an explainable AI fraud-detection backend in FastAPI with async SQLAlchemy and PostgreSQL, structured as a bounded modular monolith with per-domain service interfaces.
- Designed an idempotent, event-driven Celery + Redis pipeline (OCR → fraud analysis → forensics → risk scoring) backed by a transactional-outbox event log (Kafka in prod) for durable, replayable processing.
- Secured the platform with JWT authentication, role-based authorization, Fernet-encrypted PII, and an immutable WORM audit trail with a tamper-evident hash chain; containerized via Docker Compose with GitHub Actions CI.
- Added a NetworkX entity-relationship graph to surface shared-identity networks and fraud rings, plus a governed ML second-opinion model served with SHAP explanations and KS-test drift detection.

Flimo — *Low-Latency Semantic Search Platform*

GitHub | 2025

- Designed low-latency FastAPI services serving semantic search over vector embeddings with FAISS indexes, deployed on AWS EC2 (t3.medium) behind an Nginx reverse proxy using a multi-stage Dockerfile.
- Cut deployment time to 68 seconds with a GitHub Actions CI/CD pipeline (push → SSH → docker build → health check).
- Lowered infrastructure cost to ~\$34/month by serving HTTPS through a Cloudflare Tunnel running as an auto-restarting systemd service.
- Built a hybrid personalization engine generating composite taste vectors from user view history, filtered by rating and popularity thresholds.

FedLearn — *Browser-Native Federated Learning Platform*

GitHub | 2026

- Built scalable REST APIs supporting distributed ML training with isolated multi-tenant jobs and on-the-fly job configuration.
- Reduced tensor serialization latency from ~45 ms to under 1 ms with a zero-copy binary transport protocol replacing JSON.
- Cut network utilization 10–100× via Top-K gradient sparsification and INT8 quantization, and tolerated ~30–40% adversarial workers using Weiszfeld geometric-median aggregation.
- Implemented fault-tolerant asynchronous FedBuff aggregation with FedProx regularization for non-IID stability, plus worker-side differential privacy tracked by a Rényi DP accountant.

EDUCATION

Vellore Institute of Technology, Bhopal

Sep 2023 – May 2027

B.Tech in Computer Science and Engineering · CGPA: 8.36/10

- Relevant Coursework: Operating Systems, Computer Networks, Distributed Systems, Cloud Computing, DevOps.

ACHIEVEMENTS & CERTIFICATIONS

- Solved 200+ Data Structures and Algorithms problems, focused on backend interview preparation.
- Ranked in the Top 100 at the Zelestra x AWS ML Ascend Challenge.
- AWS Certified Cloud Practitioner (2026) and Applied Machine Learning in Python (2024); deployed production backend systems on AWS using EC2, Docker, and CI/CD.